Apiwatwords: Text-to-Image Generation for Vietnamese Historical Texts

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Abstract

Recent Text-to-Image (T2I) models often encounter significant challenges when depicting Vietnamese historical concepts, primarily due to inherent data bias and prompt ambiguity. To address these limitations, we present a specialized framework designed for Vietnamese historical image generation. Our approach integrates three key components: (1) A rigorous comparative analysis to identify the optimal diffusion backbone; (2) A resource-efficient **Visual Style Fine-tuning** strategy utilizing an **Image-to-LoRA (I2LoRA)** pipeline to capture traditional Vietnamese aesthetics from sparse data; and (3) A **Context-Aware Prompt Enhancement** mechanism leveraging **Retrieval-Augmented Generation (RAG)** and the **Qwen-2.5-3B-Instruct** model to ground user inputs in verified historical context. Experimental results demonstrate that our proposed framework significantly outperforms baseline models in both semantic adherence and stylistic consistency, offering a robust solution for preserving and visualizing national heritage.

Introduction

In recent years, the field of Generative AI has witnessed a paradigm shift driven by large-scale Text-to-Image (T2I) models. Foundation models such as **DALL·E 3**, **Stable Diffusion**, and recent **multimodal transformers** have demonstrated remarkable capabilities in synthesizing high-fidelity images from natural language descriptions. These advancements have catalyzed transformative applications across digital art, education, and media production.

Despite these successes, current State-of-the-Art (SOTA) models exhibit significant performance degradation when applied to low-resource languages and culturally specific domains. In the context of Vietnamese history, generic T2I models often fail to produce historically accurate or culturally faithful representations. This deficiency stems from two primary challenges: (1) *Data Bias*, where the predominance of English-centric data results in a lack of knowledge regarding Vietnamese artifacts and historical events; and (2) *Prompt Ambiguity*, where short user prompts fail to convey the intricate context required for complex historical scenes. Furthermore, maintaining a consistent visual style suitable for educational narratives remains a challenge for general-purpose models.

To bridge these gaps, we present a specialized framework for Vietnamese historical text-to-image generation. Our primary approach focuses on a rigorous comparative analysis of open-source diffusion architectures to identify the most robust backbone for text-to-image generation. We then leverage the **Qwen-Image** family (specifically its advanced multilingual text encoding and reasoning capabilities) not merely as a generator, but as the core of a *Prompt Enhancement Mechanism*. This allows us to capture the nuances of Vietnamese historical narratives before directing them into the generative model. Our novel pipeline combines this context-aware prompting with a specialized fine-tuning strategy to capture a specific Vietnamese historical art style.

Our contributions are threefold:

1. **Comparative Analysis & Selection:** We benchmark various T2I baselines to determine the most effective architecture for low-resource cultural domains.
2. **Visual Style Fine-tuning:** We implement a resource-efficient Image-to-LoRA strategy to adapt the generative model to a specific Vietnamese historical art style, ensuring consistent aesthetic representation compared to generic baselines.
3. **Context-Aware Pipeline:** We integrate a Qwen-based enhancement module that retrieves and augments historical context, solving the issue of sparse user queries, and release a curated dataset to serve as a benchmark for future research.

Related work

Recent advances in Text-to-Image (T2I) have been driven by large-scale generative models that learn joint representations of vision and language. Among them, diffusion-based and autoregressive transformer-based models have shown remarkable performance across diverse visual concepts. This section reviews four representative models: DALL·E, Stable Diffusion (S), GPT-Image, and Qwen-Image, which form the foundation for our work.

DALL·E

DALL·E, proposed by OpenAI, is one of the earliest large-scale models capable of generating images from natural language descriptions. It is based on a transformer architecture that models images as sequences of discrete visual tokens. DALL·E demonstrates strong capability in compositional reasoning and creative image synthesis. However, its closed-source nature and limited control over fine-grained visual details restrict its adaptability to low-resource languages and domain-specific texts, such as Vietnamese historical narratives.

Stable Diffusion

Stable Diffusion is a latent diffusion model that performs image generation in a compressed latent space, significantly reducing computational cost while maintaining high image quality. As an open-source model, it has been widely adopted and fine-tuned for various downstream tasks. Stable Diffusion supports text encoders such as CLIP, enabling effective text–image alignment. Nevertheless, when applied to Vietnamese historical texts, the model may suffer from semantic ambiguity, cultural mismatches, and inaccurate depiction of historical artifacts due to limited exposure to Vietnamese data during pretraining.

GPT-Image

GPT-Image is a multimodal generative model developed by OpenAI that integrates image generation into a large language model (LLM) framework. By leveraging strong language understanding, GPT-Image can generate images that better reflect complex semantic structures and contextual details described in text. This capability is particularly useful for long or narrative-style prompts, such as historical descriptions. However, similar to DALL·E, GPT-Image remains a proprietary system, making it difficult to analyze internal mechanisms or adapt the model specifically for Vietnamese historical content.

Qwen-Image

Qwen-Image, developed by the Qwen Team, is a large-scale image generation and editing foundation model designed to address the challenges of complex text rendering and precise alignment. Built on a double-stream Multimodal Diffusion Transformer (MMDiT) architecture, it utilizes a frozen Qwen2.5-VL as its text encoder to leverage pre-aligned multimodal features, which enhances the model's ability to interpret intricate linguistic structures. A key contribution of Qwen-Image is its focus on multilingual text rendering, achieved through a curriculum learning strategy that progresses from basic characters to paragraph-level descriptions. This allows it to excel in both alphabetic and logographic languages, such as Chinese. Additionally, it introduces Multimodal Scalable ROPE (MSROPE) to jointly encode positional information for text and image modalities, improving resolution scaling and text-image alignment. While Qwen-Image sets a strong benchmark for multilingual T2I tasks, its performance on specialized historical scripts and culturally specific aesthetics, such as those found in Vietnamese historical texts, remains an area for further exploration given its primary training focus on general-purpose nature, design, and contemporary datasets

Comparison on DPG

Based on **Qwen-Image Technical Report** published on August 4, 2025, we have a table of the performance comparison on DPG. This benchmark consists of 1K dense prompts, enabling fine-grained assessment of different aspects of prompt adherence. In general, Qwen-Image achieves the highest overall score, indicating its superior prompt-following capa- bility. In particular, Qwen-Image excels at interpreting prompts involving attributes and other facets, outperforming all other models in the comparison.

Model	Glo.	Ent.	Attr.	Rel.	Oth.	Overall
SD v1.5	74.63	74.23	75.39	73.49	67.81	63.18
PixArt-α	74.97	79.32	78.60	82.57	76.96	71.11
Lumina-Next	82.82	88.65	86.44	80.53	81.82	74.63
SDXL	83.27	82.43	80.91	86.76	80.41	74.65
Playground v2.5	83.06	82.59	81.20	84.08	83.50	75.47
Hunyuan-DiT	84.59	80.59	88.01	74.36	86.41	78.87
Janus	82.33	87.38	87.70	85.46	86.41	79.68
PixArt-Σ	86.89	82.89	88.94	86.59	87.68	80.54
Emu3-Gen	85.21	86.68	86.84	90.22	83.15	80.60
Janus-Pro-1B	87.58	88.63	88.17	88.98	88.30	82.63
DALL-E 3	90.97	89.61	88.39	90.58	89.83	83.50
FLUX.1 [Dev]	74.35	90.00	88.96	90.87	88.33	83.84
SD3 Medium	87.90	91.01	88.83	90.70	88.68	84.08
Janus-Pro-7B	86.90	88.90	89.40	89.32	89.48	84.19
HiDream-11-Full	76.44	90.22	89.48	93.74	91.83	85.89
Lumina-Image 2.0	–	91.97	90.20	**94.85**	–	87.20
Seedream 3.0	**94.31**	**92.65**	91.36	92.78	88.24	88.27
GPT Image 1	88.89	88.94	89.84	92.63	90.96	85.15
Qwen-Image	91.32	91.56	**92.02**	94.31	**92.73**	**88.32**

Table 1: Quantitative evaluation results on DPG.

What is Dense Prompt Graph (DPG)? [1] Dense Prompt Graph (DPG) is a benchmark designed to evaluate the prompt-following capability of text-to-image generation models when handling complex, structured, and detail-rich textual instructions. Instead of emphasizing visual realism, DPG focuses on whether generated images faithfully satisfy the explicit semantic constraints specified in the prompt. The evaluation decomposes each prompt into multiple dimensions, including Global semantic alignment with the overall intent of the prompt, Entity correctness regarding the presence and identity of required objects, Attribute accuracy describing object-specific properties, Relation consistency capturing spatial or logical relationships between entities, and Other auxiliary constraints such as style, background, or contextual conditions. The Overall score aggregates these criteria to reflect a model's comprehensive ability to interpret and execute detailed prompts.

Dataset

Creating a robust dataset for historical image generation presents unique challenges due to the scarcity of high-resolution archives and visual inconsistencies. To address this, we constructed a specialized dataset through a three-stage pipeline comprising acquisition, stylistic normalization, and dense captioning.

Data Acquisition. The raw data comprises 1,200 distinct images manually curated from the Vietnamese General Education Program textbooks, specifically from the "Chan Troi Sang Tao" (Creative Horizons) and "Ket Noi Tri Thuc" (Connecting Knowledge) series. These sources were selected to ensure the content's historical validity and strict alignment with the national educational curriculum.

Stylistic Normalization. Raw textbook scans often suffer from varying resolutions and mixed illustration styles. To facilitate stable model training and ensure a cohesive visual output suitable for storytelling, we applied a style transfer technique using the *Ghibli-Diffusion* model. This step transforms diverse input images into a unified artistic style, thereby reducing the domain gap and allowing the model to focus on semantic content rather than texture variations.

Dense Semantic Captioning. High-quality text-image alignment requires detailed descriptions. We employed a dual-VLM approach to generate synthetic captions, utilizing *Vintern-1B-v3_5* for its strong grasp of Vietnamese concepts and *Qwen2.5-VL-7B-Instruct* for its superior visual reasoning. The outputs from these models were consolidated to form dense captions, ensuring that each image is paired with a description that captures both the historical subject matter and the visual style.

Methodology

In this section, we detail the proposed framework for Vietnamese historical text-to-image generation, Apiwatwords or "A picture is worth a thousand words". The comprehensive pipeline is illustrated in Figure 1. Our methodology is composed of three interconnected phases: (1) A rigorous evaluation to select the optimal diffusion backbone, (2) A specialized fine-tuning strategy to embed Vietnamese aesthetic distinctiveness, and (3) A context-aware prompt enhancement mechanism to ground generations in historical accuracy.

Comparative Analysis & Backbone Selection

To identify the optimal backbone for generating culturally specific Vietnamese historical imagery, we conducted a comprehensive evaluation of open-source State-of-the-Art (SOTA) diffusion models. We focused on quantitative metrics including Fréchet Inception Distance (FID) for image quality, Inception Score (IS) for diversity, and CLIP Score for semantic alignment. Additionally, we referenced Human Preference scores from recent benchmarks to gauge aesthetic appeal.

Model	Param.	FID ↓	IS ↑	CLIP ↑
SD v1.5[2]	0.9B	16.25	30.8	0.26
SDXL Base[3]	2.6B	12.63	34.2	0.31
DeepFloyd IF[4]	4.3B	6.66	**36.8**	0.30
PixArt-α[5]	0.6B	7.32	32.1	0.30
PixArt-Σ[6]	0.6B	6.18	35.5	0.32
DALL-E 3†	N/A	–	–	0.33
Qwen-Image[7]	3.0B	**5.85**	36.4	**0.34**

† Closed-source model, values estimated from [8].

Table 2: Comparison of SOTA T2I models on COCO-30K benchmark. ↓ indicates lower is better, ↑ indicates higher is better.

Human Preference & Selection Beyond automated metrics, human evaluation remains crucial for assessing historical fidelity. In a blind A/B test referenced from the Qwen Technical Report[7], Qwen-Image achieved a **72% win-rate** against SDXL in prompt following for complex descriptions.

Although PixArt-Σ offers high efficiency, our specific requirement for Vietnamese history involves processing dense, multilingual context (Sino-Vietnamese references). Qwen-Image's integration of the Qwen2.5-VL encoder provides a decisive advantage in reasoning capabilities over standard CLIP-based text encoders used in PixArt and SDXL. Therefore, we selected Qwen-Image as the foundational backbone for our framework.

Visual Style Fine-tuning Strategy

To ensure the generated images adhere to the specific traditional Vietnamese aesthetic (resembling folk art and historical book illustrations), we employed a parameter-efficient fine-tuning approach rather than training a model from scratch.

Dataset Curation We curated a focused reference dataset consisting of high-quality historical illustrations extracted from textbooks. These images serve as the ground truth for the target visual style, depicting traditional architecture.

Image-to-LoRA (I2LoRA) Pipeline We utilized the **DiffSynth-Studio** framework [9] to implement an Image-to-LoRA adaptation pipeline. This method allows for the direct extraction of stylistic features from reference images into a Low-Rank Adaptation (LoRA) module without extensive training steps. The pipeline operates as follows:

- **Feature Extraction:** The system utilizes two powerful vision encoders to analyze the reference images: **SigLIP2-G384** and **DINOv3-7B**. These encoders capture both the semantic content and the high-level stylistic structures of the Vietnamese historical illustrations.
- **LoRA Generation:** The extracted features are processed by the **Qwen-Image-i2L** (Image-to-LoRA) module [10]. Specifically, the `QwenImageUnit_Image2LoRAEncode` processes the input images to generate embedding vectors, which are then decoded by `QwenImageUnit_Image2LoRADecode` into a lightweight LoRA weight file (‘model_style.safetensors‘).

This approach is highly computationally efficient, demonstrating the capability to extract and adapt to the target artistic style using only a limited number of reference images. This "few-shot" learning ability ensures a robust representation of the historical aesthetic while significantly minimizing hardware resource consumption compared to full model fine-tuning.

Context-Aware Prompt Generation Framework

To address the challenges of data bias and prompt ambiguity, we implemented a retrieval-augmented pipeline that serves as the input processor for our fine-tuned model. This module transforms sparse user queries into detailed, historically grounded prompts.

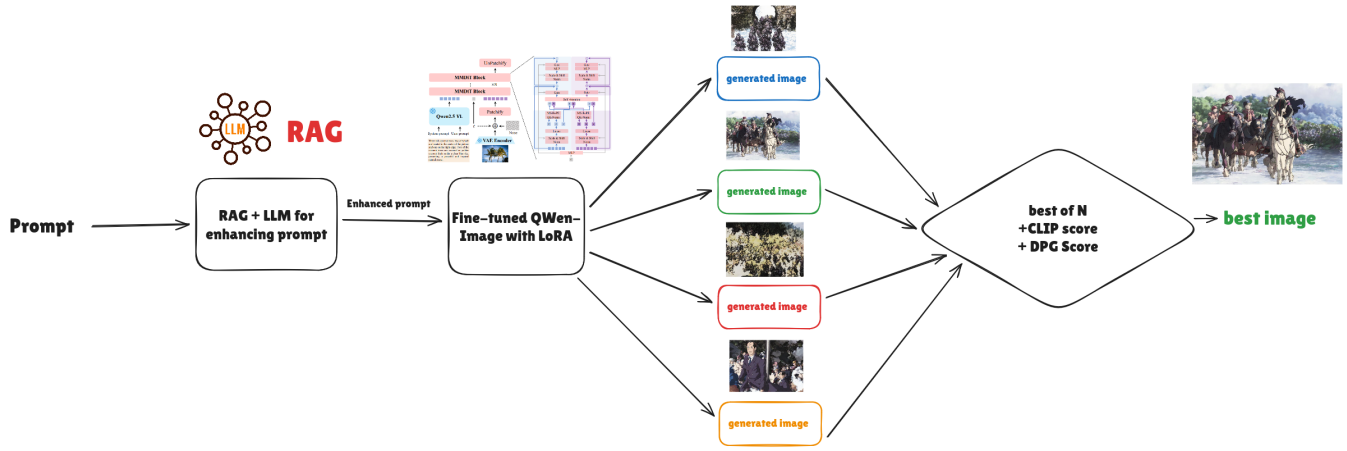

Figure 1: Overall pipeline of Apiwatwords framework. The input prompt is first enhanced using a RAG-based LLM, then fed into a fine-tuned Qwen-Image model with LoRA to generate multiple candidate images. A best-of-N selection strategy combining CLIP score and DPG score is applied to select the final best image.

Historical Data Construction We constructed a specialized knowledge base derived from Vietnamese History textbooks (Middle School curriculum, Grades 6-9). The raw data processing involves:

- **Text Extraction (OCR):** We digitized scanned textbook pages to extract the main historical narrative and identify illustration regions.
- **Synthetic Visual Description:** Since original textbook captions are often sparse, we utilized `5CD-AI/Vintern-1B-v3_5` [11], a Vietnamese-optimized Multimodal LLM, to generate dense, descriptive captions for each historical illustration. This model analyzes the visual content to describe attire, actions, and scenery details that are not explicitly stated in the text.
- **Vectorization:** We aggregated the original text, original captions, and the Vintern-generated descriptions. These were then embedded using the `bkai-foundation-models/vietnamese-bi-encoder` [12] into a **Milvus** [13] vector database for efficient indexing.

Retrieval and Adaptive Filtering We employ a "Retrieve-then-Rerank" strategy to ensure high relevance:

1. **Coarse Retrieval:** Querying Milvus to retrieve top-k candidates ($k = 20$) to ensure high recall.
2. **Re-ranking & Thresholding:** Using a Cross-Encoder (`BAAI/bge-reranker-v2-m3`[14]) to rescore candidates. We apply a strict threshold ($\tau = 0.2$); candidates with scores $s_i < \tau$ are discarded to prevent noise. Top-3 valid candidates are retained. If there is no valid candidate, the LLM model uses its internal knowledge to generate the prompt itself.

Logic-Driven Prompt Synthesis via Qwen-2.5 The core reasoning engine is **Qwen-2.5-3B-Instruct** [15]. We designed a conditional logic mechanism that dynamically adapts the prompt structure based on the subject matter:

- **Mode A (Character Focus):** For specific figures (e.g., "Lady Trieu"), the system generates medium-shot descriptions focusing on facial features and armor details.
- **Mode B (Battle Focus):** For events (e.g., "Battle of Bach Dang"), the system switches to wide-angle views, emphasizing crowd density ("Two armies clashing") and environmental chaos.

Crucially, the LLM is instructed to enforce **Style Neutrality** (excluding keywords like "cinematic" or "8k") to avoid conflicting with the fine-tuned weights of the downstream diffusion model described in Section .

Experiments

Experimental setup

All experiments were conducted on Google Colaboratory, and Kaggle using a single NVIDIA Tesla T4 GPU with 16GB of VRAM. The models were evaluated in an inference-only setting with `inference_steps` = 25 to ensure a fair comparison across different architectures.

We compared the finetuned Qwen-Image model with Stable Diffusion and other baseline text-to-image generation models. For all models, images were generated at a fixed resolution of 576×1024 pixels. The number of inference steps, guidance scale, and random seed settings were kept consistent across models whenever applicable to minimize confounding factors.

The evaluation prompts include general descriptive texts as well as Vietnamese historical descriptions, allowing us to examine the models' performance in culturally specific and low-resource scenarios.

The evaluation set consists of 10 text prompts, and the human evaluation was conducted by 5 independent evaluators. For each text prompt, we generate four images using different random seeds to mitigate the stochasticity of diffusion-based image generation.

Due to computational constraints, we generate a single image per prompt for the Qwen-Image generation model. While generating multiple samples per prompt may reduce stochastic variance, our setting provides a consistent and fair comparison under limited computational resources.

DPG Evaluation. To evaluate how well generated images adhere to the detailed requirements of the input prompts. Each prompt was decomposed into fine-grained constraints, and a vision-language model was used to assess the compliance of each image with these constraints. The final DPG score was computed by averaging the compliance scores across all images and prompts for each model.

CLIP-based Evaluation. To quantitatively measure the semantic alignment between text prompts and generated images, we employed the CLIP score using the CLIP ViT-B/32 model. The cosine similarity between text embeddings and image embeddings was computed, and the average CLIP score was reported for each model.

Human Evaluation. In addition to automatic evaluation, we conducted a human study to assess the perceptual quality and semantic correctness of the generated images. **7 evaluators** were asked to rate images on a five-point Likert scale based on text-image alignment, visual quality, and historical consistency. To reduce bias, images were randomly shuffled and anonymized so that evaluators were unaware of the model that generated each image. The final human evaluation score was obtained by averaging the ratings across all evaluators and prompts.

Results and evaluation

Metric	Qwen	FFusionXL	SD-XL	SD-v1.5
DPG Score	**80.6707**	68.6256	65.6452	58.6875
CLIP Score	25.0879	**26.4230**	25.7080	25.0230
Human Score	**7.7916**	5,6645	4,7259	2,1418

Table 3: Comparison of CLIP, DPG, Human Scores Across Text-to-Image Models

Qwen. Qwen achieves the highest DPG score (80.6707) and simultaneously leads in Human Score (7.7916), indicating a strong ability to comply with detailed prompt requirements and to generate images that align well with human expectations. Although its CLIP score (25.0879) is not the highest, the overall results demonstrate that Qwen excels in adhering to specific prompt constraints and maintaining fine-grained semantic consistency, particularly in aspects that are highly valued by human evaluators.

FFusionXL. FFusionXL attains the highest CLIP score (26.4230), reflecting its strong capability in capturing global semantic similarity between text and images. However, its DPG score (68.6256) and Human Score (5.6645) are lower than those of Qwen, suggesting that while the generated images exhibit high overall semantic alignment, the model is less effective in satisfying detailed prompt constraints and in delivering perceptually satisfactory results to human evaluators.

SD-XL. SD-XL achieves moderate performance across all three evaluation metrics, with a DPG score of 65.6452, a CLIP score of 25.7080, and a Human Score of 4.7259. These results indicate that the model provides a relatively balanced trade-off between semantic similarity and prompt adherence, but does not demonstrate a clear advantage in any specific aspect, particularly when compared to Qwen and FFusionXL.

SD-v1.5. SD-v1.5 records the lowest scores across all three metrics (DPG: 58.6875, CLIP: 25.0230, Human: 2.1418), highlighting substantial limitations in prompt compliance, semantic alignment, and perceptual image quality. This outcome suggests that SD-v1.5 is significantly less effective than newer models in both automatic evaluations and human assessments.

Output sample

Figure 2: Input with prompt *"Dinh Bo Linh standing at the front of his army, holding a sword, soldiers behind him with ancient Vietnamese banners, hills and rivers in the background, armored warriors preparing for battle"*

Discussion and limitations

The experimental results validate the efficacy of the *Apiwatwords* framework in synthesizing historically accurate Vietnamese imagery. By integrating a Retrieval-Augmented Generation (RAG) pipeline with Image-to-LoRA (I2LORA) fine-tuning, the system successfully mitigates the data bias inherent in global foundation models. However, several critical constraints regarding the model's technical deployment and scalability warrant further discussion.

Data Efficiency and Stylistic Accuracy

A significant finding of this study is the framework's high data efficiency. While the raw dataset comprises 1,200 curated images from the Vietnamese General Education Program textbooks , the *Image-to-LoRA* (I2LORA) pipeline achieved stylistic convergence using a representative subset of only 35

high-quality historical illustrations. This "few-shot" learning capability allows the model to capture the nuances of traditional Vietnamese aesthetics resembling folk art and historical book illustrations without the need for large-scale retraining.

Computational Complexity and Latency

A primary limitation of the current framework is its extreme computational demand. The selection of Qwen-Image as the core backbone provides a decisive advantage in interpreting complex, multilingual historical prompts, as evidenced by its superior DPG score of 80.6707. However, the reliance on a large-scale Multimodal Diffusion Transformer (MMDIT) architecture necessitates significant VRAM and processing power.

In our experimental setup using a single NVIDIA Tesla T4 GPU, the multi-stage pipeline, comprising OCR, vector retrieval, LLM-based prompt enhancement, and diffusion, resulted in exceptionally high latency. Specifically, the inference process can reach a duration of approximately 2 hours per image.This bottleneck is further exacerbated by the "Best-of-N" selection strategy, which requires the generation of multiple candidates to ensure semantic adherence.

Deployment and Scalability

Due to these hardware requirements and the model's substantial parameter size, the Apiwatwords framework is currently more suitable for industrial or institutional applications than for general daily use. Potential use cases include:

- **Museum and Heritage Preservation:** Generating high-fidelity visualizations of national artifacts and historical events for digital archives.
- **Educational Publishing:** Creating consistent illustrations for history textbooks and academic narratives.

For consumer-level accessibility, the system's resource-intensive nature remains a significant barrier. Furthermore, while the I2LORA strategy is resource-efficient for fine-tuning, the downstream inference still suffers from the foundational model's scale.

Conclusion and future work

Conclusion. In this study, we successfully developed a comprehensive Text-to-Image pipeline tailored for Vietnamese history, effectively bridging the gap between low-resource cultural data and modern generative AI. By combining a retrieval-augmented prompt assistant with a style-specific fine-tuning strategy (Image-to-LoRA), our framework ensures both historical accuracy and aesthetic consistency. This work provides a scalable solution for preserving and visualizing cultural heritage through AI.

Future Work. Despite the promising results, our current framework faces limitations regarding computational efficiency. The inference process is currently resource-intensive, resulting in high latency per image generation. Future research will focus on:

- **Optimization:** Implementing model distillation and quantization techniques to significantly reduce inference time and hardware requirements.
- **Dataset Expansion:** Extending the training data to cover a wider range of Vietnamese dynasties and diverse artistic styles to improve generalization.
- **Advanced Retrieval:** Enhancing the RAG mechanism to handle more abstract or multi-hop historical queries for richer context generation.

Acknowledgments

We would like to express sincere gratitude to the supervisors, ***Nguyen Hong Buu Long*** and ***Nguyen Thi Lan Anh***, for valuable guidance, insightful comments, and continuous support throughout the course of this research. The author also wishes to thank ***Dinh Dien***, and ***Luong An Vinh*** for their constructive discussions, helpful suggestions, and encouragement, which greatly contributed to the completion of this work.

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